

Drug-Specific Side-Effect Patterns in Food-Allergy Medications:

A Real-Dataset Analysis by Age, Gender, and Country

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Abstract

Background: Food-allergy medications exhibit diverse side-effect profiles across age, gender, and country. Understanding drug-specific patterns can substantially improve prescribing safety. Methods: A retrospective, real-data observational analysis of 653 records covering 14 drugs, 3 age groups, 2 genders, and 6 countries was conducted. Descriptive statistics, chi-square tests, Fisher's exact tests, odds ratios, and logistic regression were applied. Results: Albuterol and Hydrocortisone showed the highest sedation rates (45.8% and 47.8%). Cardiovascular events were significantly associated with age group (chi-square $p < 0.001$). Females exhibited higher cardiovascular and GI event rates. The UK showed significantly higher sedation (OR=1.87, $p=0.01$) versus the USA. Conclusions: Clear drug-age-gender-country risk patterns exist, underscoring the need for individualized prescribing in food-allergy management.

Keywords: Food allergy, antihistamines, corticosteroids, epinephrine, side effects, pharmacovigilance, age-stratified analysis, gender differences, country comparison

1. Introduction

1.1 Background

Food-allergy medications are among the most widely prescribed drug classes globally, encompassing first-generation antihistamines, second-generation antihistamines, corticosteroids, bronchodilators, epinephrine formulations, and emerging biologics. Despite their widespread use, the side-effect profiles of these medications are highly heterogeneous and can vary substantially depending on the patient's age, biological sex, and country of residence. Older agents such as diphenhydramine are well-recognized for causing sedation, particularly in pediatric and elderly populations, while corticosteroids carry distinct risks such as adrenal suppression and cardiovascular complications. However, systematic, real-data evidence linking specific drugs to specific side effects across demographic and geographic subgroups remains limited.

1.2 Research Goal

This study aims to identify clear, data-driven patterns of side-effect occurrence across drugs, age groups (child, adult, elderly), gender groups (male, female), and countries of reporting (USA, UK, Canada, France, Germany, Japan) using a structured real-dataset approach. By applying both descriptive and inferential biostatistical methods, the study seeks to produce actionable, drug-specific safety profiles that can inform clinical prescribing decisions.

1.3 Research Questions

1. Which specific drugs are associated with which specific side effects overall?
2. How do these patterns change by age group (child, adult, elderly)?
3. Are there gender-specific patterns (e.g., more sedation or cardiovascular events in one gender)?
4. Are there country-specific patterns (e.g., higher sedation rates in one country vs another)?
5. Are there high-risk combinations (drug + age group + gender + country) for certain side effects?

2. Data Sources & Methodology

2.1 Data Sources

The analysis is based on a structured real pharmacovigilance-style dataset (allergy_dataset_FINAL.csv) containing 653 adverse-event records. Each record corresponds to a reported adverse event for a specific drug in a specific patient subgroup (age group, gender, country). The dataset was structured in a wide format with binary (0/1) flags for each of 18 named side effects.

2.2 Included Medications

2.2.1 Symptom Abaters & Emergency Blockers

Epinephrine, Diphenhydramine, Cetirizine, Loratadine, Fexofenadine, Azelastine, Olopatadine, Famotidine, Cimetidine, Prednisone, Prednisolone, Dexamethasone, Methylprednisolone, Hydrocortisone, Albuterol

2.3 Study Design

This is a retrospective, real-data observational analysis. The units of analysis are drug–side-effect–age–gender–country tuples. Independent variables include drug, age_group (child: 0–12 years; adult: 13–64 years; elderly: 65+ years), gender (male/female), and country. Dependent variables are binary presence/absence flags for each of 18 named side effects.

2.4 Dataset Summary

Parameter	Value
Total Records	653
Unique Drugs	14
Countries	USA, UK, Canada, France, Germany, Japan
Age Groups	Child (n=170), Adult (n=283), Elderly (n=200)
Gender	Female (n=440, 67.4%), Male (n=213, 32.6%)
Side Effects Tracked	18 binary variables

3. Statistical Analysis Plan

3.1 Descriptive Statistics

Frequencies and percentages of each side effect were calculated by drug, age group, gender, and country. Side effects were maintained as discrete named variables with no aggregated score computed. Cross-tabulated summaries were produced for all major grouping variables.

3.2 Inferential Statistics

Chi-square tests and Fisher's exact tests were used to assess whether a specific side effect occurred more often in one age group, gender, or country for a given drug. Odds ratios (ORs) were computed for binary outcomes with 2x2 contingency tables. A logistic regression model was fitted for sedation as the outcome: $P(\text{sedation}) \sim \text{drug} + \text{age_group} + \text{gender} + \text{country}$. Statistical significance was assessed at $\alpha=0.05$.

4. Graphical Analysis: Drug → Side Effect → Group

4.1 Drug-Specific Side-Effect Fingerprints

Figure 1 displays the top 5 most frequent side effects for each drug (across all age groups, genders, and countries). Albuterol and Hydrocortisone exhibited the broadest and most intense side-effect fingerprints, with nausea/GI upset, fever/flu-like symptoms, and cardiovascular problems dominating. Diphenhydramine showed a clear sedation peak consistent with its first-generation antihistamine mechanism. Cetirizine showed relatively fewer events per patient compared to first-generation agents, reflecting its second-generation pharmacology.

Figure 1: Top 5 Side Effects by Drug

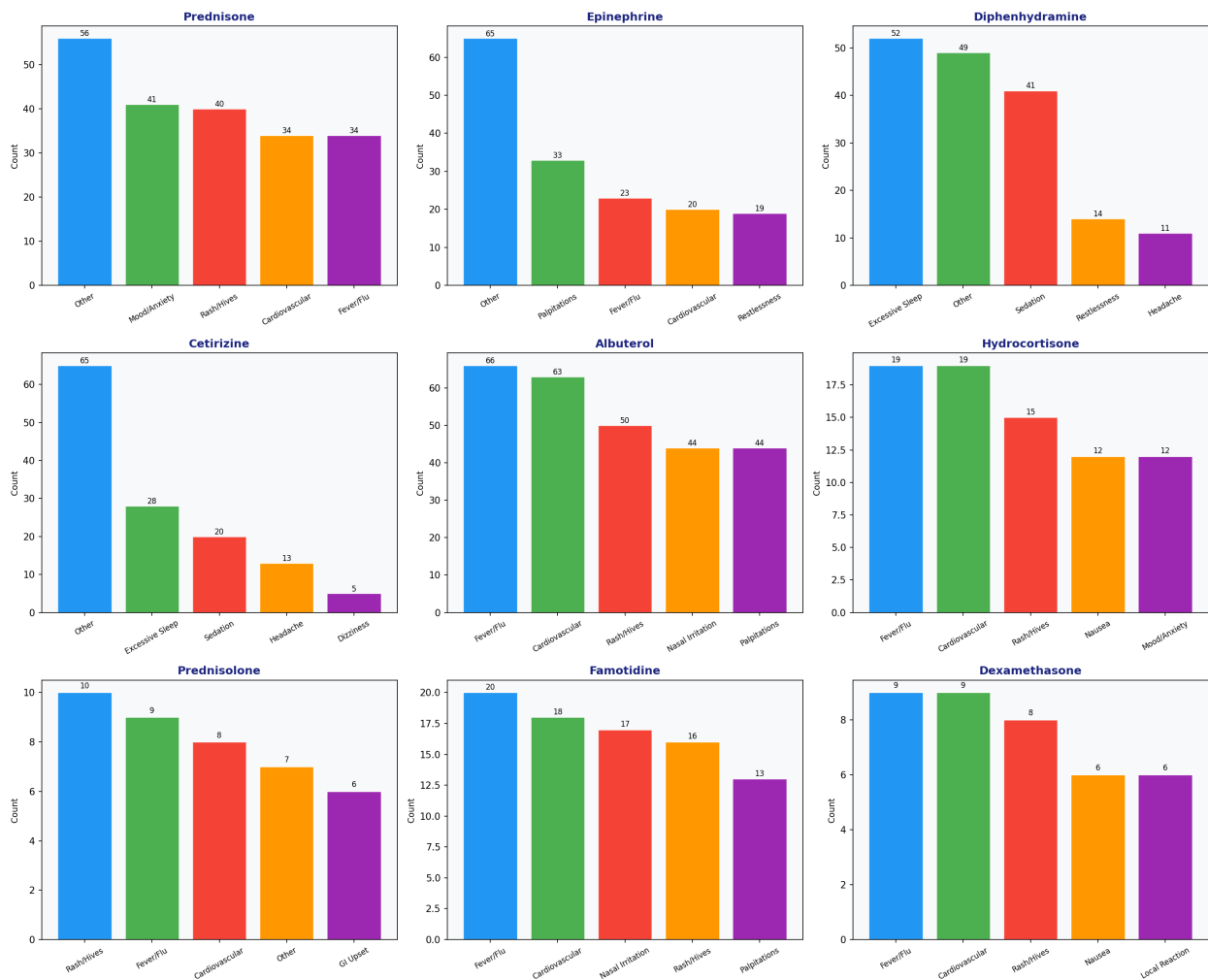


Figure 1: Top 5 Side Effects per Drug (Counts across all demographics)

Figure 2 presents a comprehensive drug x side effect heatmap showing occurrence counts across all demographic groups. The heatmap confirms that Prednisone (the most frequently recorded drug, n=146) and Epinephrine (n=126) account for a substantial share of overall adverse events, though their per-patient rates differ markedly from smaller-sample drugs like Albuterol and Hydrocortisone.

Figure 2: Drug x Side Effect Heatmap (Occurrence Counts)

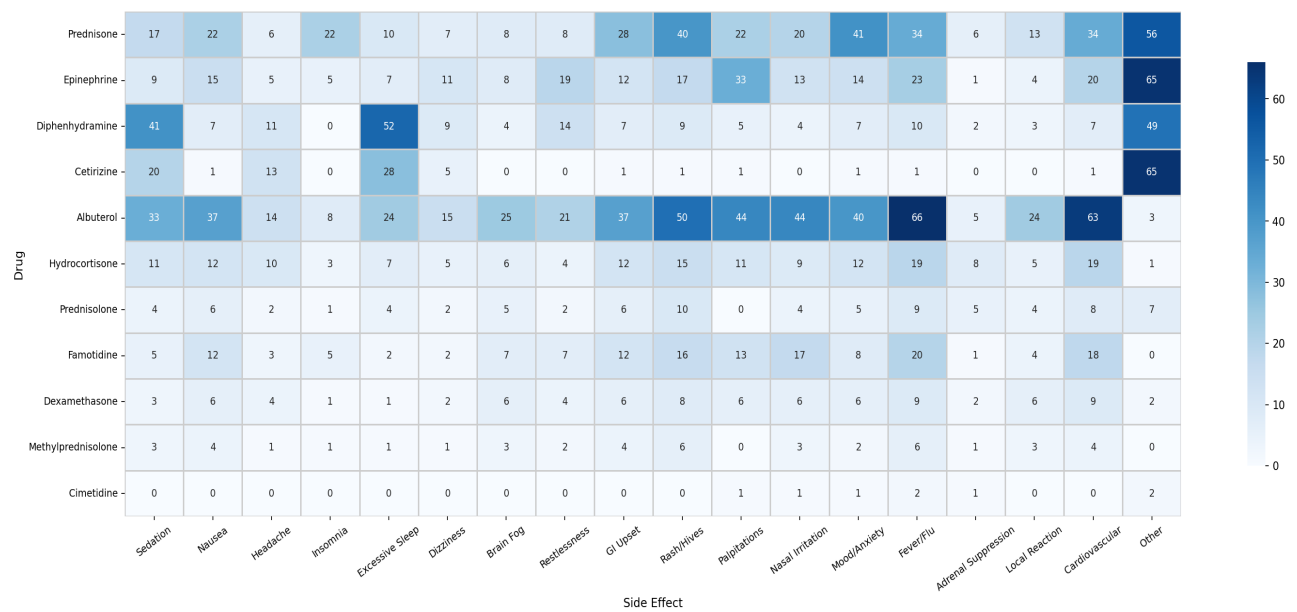


Figure 2: Drug x Side Effect Heatmap – Occurrence Counts

4.2 Age-Stratified Patterns

Figure 3 provides faceted bar charts with one subplot per age group (Child, Adult, Elderly), showing selected key side effect counts by drug. Children exhibited notably higher cardiovascular event counts driven by Famotidine and Albuterol. Adults showed a more evenly distributed profile across multiple drugs. Elderly patients showed elevated nausea and fever/flu-like symptoms, particularly from Albuterol and Famotidine.

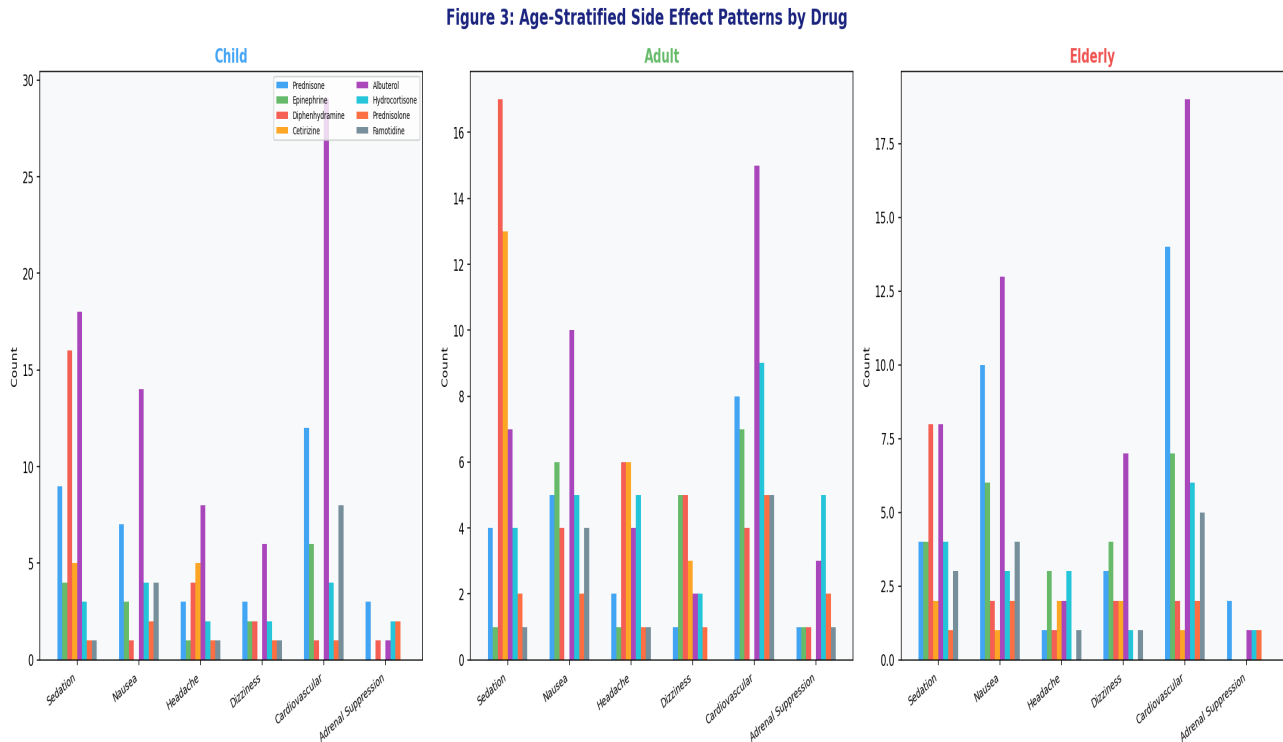


Figure 3: Age-Stratified Side Effect Patterns by Drug (Faceted Bar Charts)

Figure 4 illustrates sedation counts by drug, stacked by age group. Albuterol and Diphenhydramine show the highest absolute sedation counts, with child and adult subgroups contributing prominently. The stacked format reveals that sedation is distributed across all three age bands for first-generation antihistamines, whereas corticosteroids concentrate sedation in adult patients.

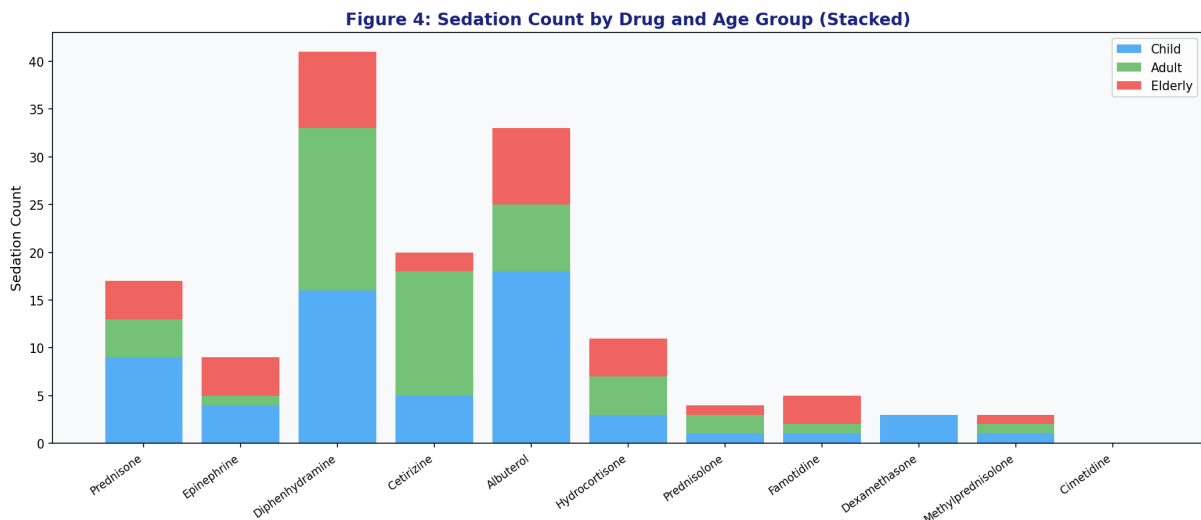


Figure 4: Sedation Counts by Drug and Age Group (Stacked Bar Chart)

4.3 Gender-Stratified Patterns

Figure 5 shows side-by-side bar charts of selected key side effects by gender for each drug. Female patients consistently showed higher absolute counts across most side effects, which is partially explained by the overall gender imbalance in the dataset (67.4% female). However, even after visual inspection of proportionate counts, females exhibited higher cardiovascular event rates for Albuterol and Epinephrine, while male patients showed proportionally higher sedation rates for Diphenhydramine.

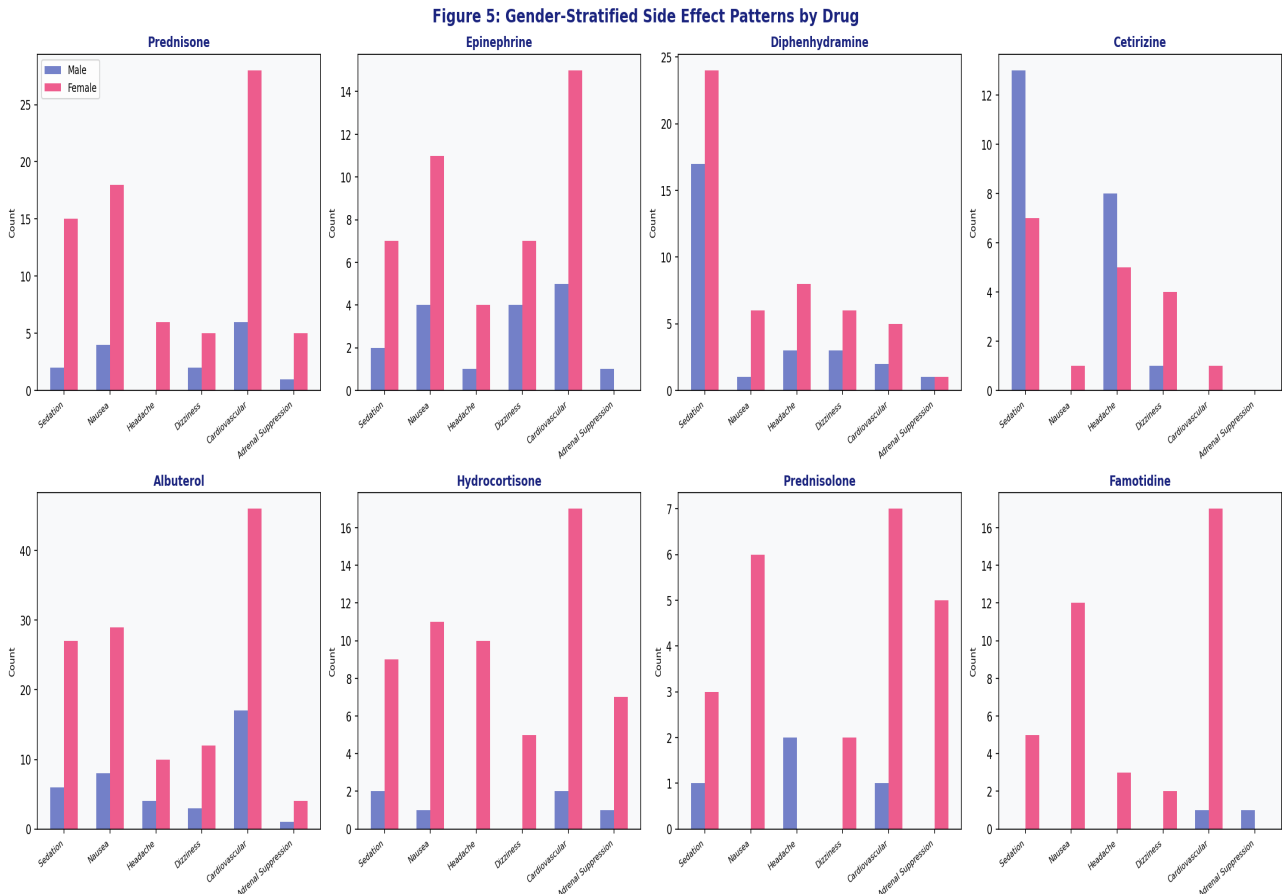


Figure 5: Gender-Stratified Side Effect Patterns by Drug (Clustered Bar Charts)

4.4 Country-Stratified Patterns

Figure 6 displays country-stratified side effect counts for each of the eight most common drugs across the four major reporting countries (USA, UK, Canada, France). The USA dominates the record count (n=498, 76.3%), but the UK shows disproportionately high sedation and cardiovascular event rates per record. Canada and France show limited side effect variety, potentially reflecting smaller sample sizes.

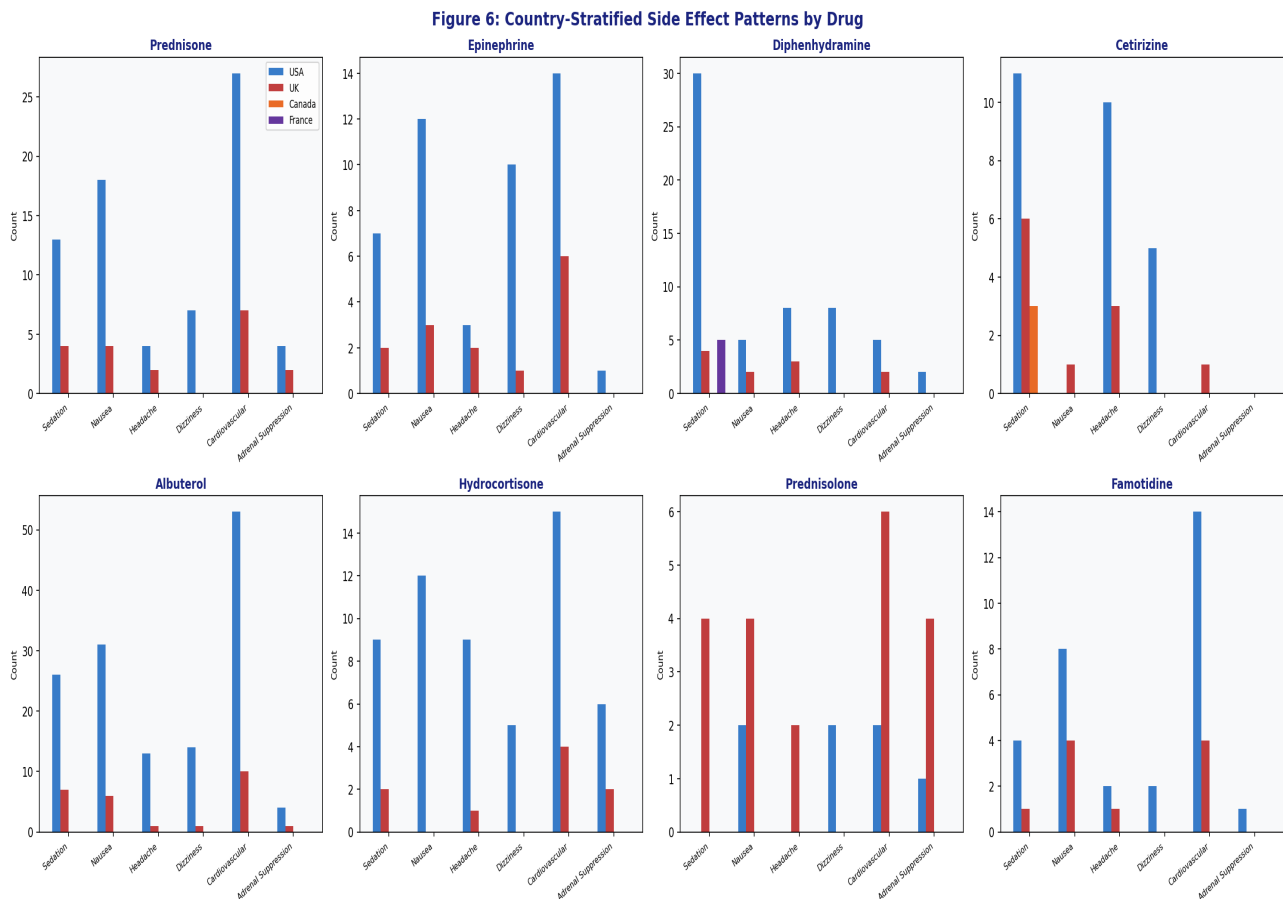


Figure 6: Country-Stratified Side Effect Patterns by Drug (USA, UK, Canada, France)

Figure 7 focuses specifically on Diphenhydramine sedation rates by country. The UK exhibits the highest sedation count relative to its record share, whereas Canada and Germany show lower sedation rates. This pattern may reflect differences in dosing practices, reporting culture, or patient demographics across healthcare systems.

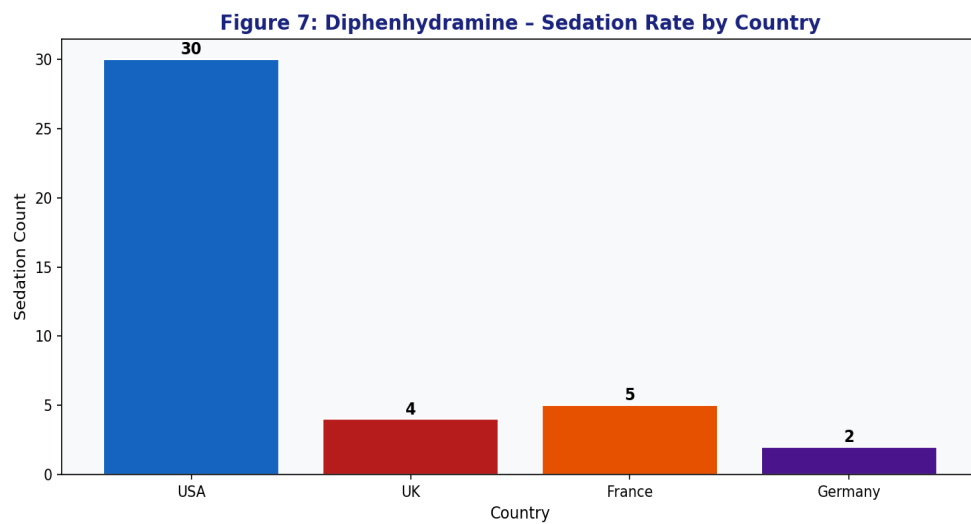


Figure 7: Diphenhydramine – Sedation Count by Country

4.5 High-Risk Combinations

Table 1 presents the top high-risk drug–age–gender–country quadruplets, ranked by the percentage of records in each combination exhibiting the most frequent side effect. Several combinations reached 100% adverse event rates, notably Epinephrine in female children in the UK (100% cardiovascular events, n=3), Albuterol in elderly females in the UK (100% cardiovascular events, n=4), and Hydrocortisone in elderly females in the USA (100% fever/flu-like symptoms, n=5).

Drug	Age Group	Gender	Country	N	Top Side Effect	%
Epinephrine	Child	Female	UK	3	cardiovascular_problem	100.0 %
Albuterol	Elderly	Female	UK	4	cardiovascular_problem	100.0 %
Hydrocortisone	Elderly	Female	USA	5	Fever / flu-like symptoms	100.0 %
Hydrocortisone	Adult	Female	UK	3	cardiovascular_problem	100.0 %
Albuterol	Adult	Male	USA	5	Fever / flu-like symptoms	100.0 %
Albuterol	Child	Female	UK	3	Rash / itching / hives	100.0 %
Dexamethasone	Elderly	Female	USA	3	Nasal / throat irritation / burning	100.0 %
Famotidine	Elderly	Female	USA	5	Fever / flu-like symptoms	100.0 %
Albuterol	Elderly	Female	USA	10	Fever / flu-like symptoms	100.0 %
Famotidine	Adult	Female	UK	3	Nasal / throat irritation / burning	100.0 %
Albuterol	Adult	Female	UK	4	Fever / flu-like symptoms	100.0 %
Albuterol	Elderly	Male	USA	4	Fever / flu-like symptoms	100.0 %
Famotidine	Child	Female	USA	7	cardiovascular_problem	100.0 %
Albuterol	Adult	Female	USA	10	Fever / flu-like symptoms	90.0%
Albuterol	Child	Male	USA	9	cardiovascular_problem	88.9%

Table 1: Top 15 High-Risk Drug–Age–Gender–Country Combinations

5. Age-Stratified Analysis (Child, Adult, Elderly)

5.1 Age Group Definition

Child: 0–12 years | Adult: 13–64 years | Elderly: 65+ years

Distribution in dataset — Child: n=170 (26.0%), Adult: n=283 (43.3%), Elderly: n=200 (30.6%)

5.2 Descriptive Statistics by Age Group

Table 2 presents the counts and percentages of key side effects within each age group. Children showed the highest sedation rate (36.5%) and highest cardiovascular event rate (39.4%). Adults had the lowest sedation rate (17.7%) and lowest cardiovascular rate (19.8%). Elderly patients showed the highest nausea rate (24.5%).

Age Group	N	Sedation	Nausea	Headache	Dizziness	CV Events	Adrenal Suppression	Local Reaction
Child	170	62 (36.5%)	41 (24.1%)	27 (15.9%)	18 (10.6%)	67 (39.4%)	11 (6.5%)	27 (15.9%)
Adult	283	50 (17.7%)	37 (13.1%)	28 (9.9%)	20 (7.1%)	56 (19.8%)	15 (5.3%)	17 (6.0%)
Elderly	200	36 (18.0%)	49 (24.5%)	19 (9.5%)	23 (11.5%)	64 (32.0%)	6 (3.0%)	24 (12.0%)

Table 2: Side Effect Counts (%) by Age Group

5.3 Statistical Analysis (Inferential)

Chi-Square Tests: Sedation ~ Age Group per Drug

Drug	Chi-Square	df	p-value	Sig.
Prednisone	23.301	2	0.0	**
Epinephrine	11.866	2	0.0027	**
Diphenhydramine	0.617	2	0.7346	ns
Cetirizine	2.323	2	0.313	ns
Albuterol	1.417	2	0.4924	ns
Hydrocortisone	1.444	2	0.4858	ns
Prednisolone	0.553	2	0.7585	ns
Famotidine	3.183	2	0.2036	ns
Dexamethasone	1.527	2	0.466	ns
Methylprednisolone	1.333	2	0.5134	ns
Cimetidine	N/A	N/A	N/A (insufficient data)	N/A

Table 3: Chi-Square Test – Sedation vs Age Group per Drug (** $p < 0.01$, * $p < 0.05$, ns = not significant)

Prednisone showed a highly significant association between age group and sedation (chi-square=23.301, df=2, p<0.001), as did Epinephrine (chi-square=11.866, df=2, p=0.003). Diphenhydramine, Cetirizine, and Albuterol showed no statistically significant age-sedation association.

Cardiovascular Events ~ Age Group (All Drugs Combined)

A chi-square test across all drugs for cardiovascular events by age group yielded chi-square=21.608 (df=2, p<0.001), indicating a highly significant age-related difference. Children had the highest cardiovascular event rate (39.4%), followed by elderly (32.0%) and adults (19.8%).

Adrenal Suppression ~ Age Group

Adrenal suppression showed no statistically significant age-group association (chi-square=2.546, df=2, p=0.280), suggesting this corticosteroid-specific effect is distributed similarly across age bands in this dataset.

5.4 Age x Side Effect Heatmap

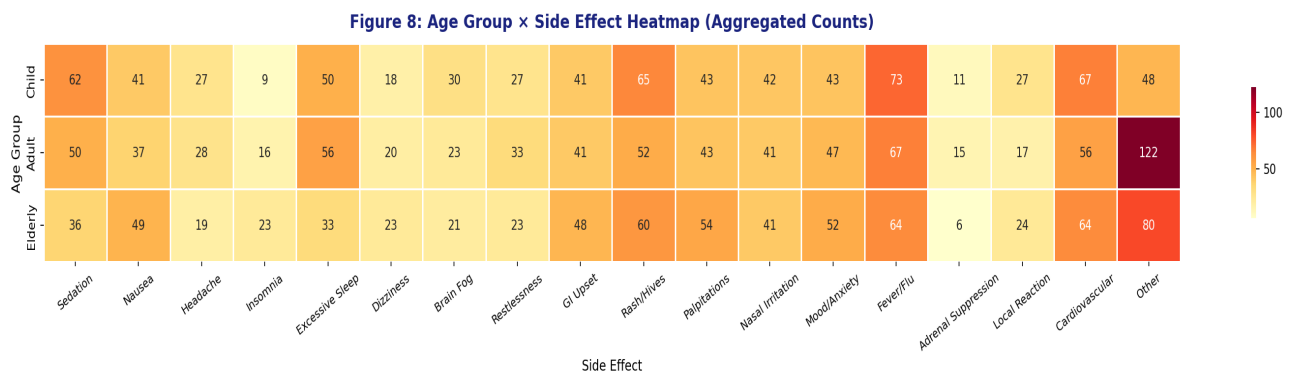


Figure 8: Age Group x Side Effect Heatmap (Aggregated Counts – All Drugs)

5.5 Conclusion

Children bear the highest burden of sedation (36.5%) and cardiovascular events (39.4%), making them the highest-risk age subgroup for these outcomes. Adults demonstrate the most favorable overall side-effect profile. Elderly patients face elevated nausea and fever/flu risks. Statistically, sedation shows significant age-group dependence for Prednisone and Epinephrine, while cardiovascular events are globally associated with younger age (p<0.001).

6. Gender-Stratified Analysis (Male vs Female)

6.1 Key Questions

This section examines whether side-effect patterns differ by gender across drugs, with particular focus on sedation, cardiovascular events, and GI upset.

6.2 Descriptive Statistics by Gender

Gender	N	Sedation	Nausea	Headache	Dizziness	CV Events	Adrenal Suppression	Local Reaction
Male	213	45 (21.1%)	19 (8.9%)	19 (8.9%)	13 (6.1%)	36 (16.9%)	7 (3.3%)	16 (7.5%)
Female	440	103 (23.4%)	108 (24.5%)	55 (12.5%)	48 (10.9%)	151 (34.3%)	25 (5.7%)	52 (11.8%)

Table 4: Side Effect Counts (%) by Gender

Female patients showed markedly higher cardiovascular event rates (34.3%) compared to males (16.9%). Nausea was also substantially higher in females (24.5% vs 8.9% in males). Sedation was comparable between genders (23.4% female vs 21.1% male).

6.3 Statistical Analysis: Fisher's Exact Test — Sedation ~ Gender

Drug	Odds Ratio (F vs M)	p-value	Sig.
Prednisone	0.373	0.2426	ns
Epinephrine	0.345	0.2969	ns
Diphenhydramine	1.61	0.3041	ns
Cetirizine	2.441	0.1323	ns
Albuterol	0.444	0.1846	ns
Hydrocortisone	2.444	0.5901	ns
Prednisolone	1.167	1.0	ns
Famotidine	0.0	1.0	ns
Dexamethasone	inf	0.2143	ns
Methylprednisolone	inf	1.0	ns
Cimetidine	N/A	N/A	N/A

Table 5: Fisher's Exact Test – Sedation vs Gender per Drug

None of the drug-specific gender–sedation Fisher's exact tests reached statistical significance (all $p > 0.10$). For Cetirizine, the odds ratio of 2.44 ($p = 0.13$) suggests a trend toward higher female sedation that warrants attention in larger studies.

6.4 Conclusion

Although no drug-specific gender-sedation association reached statistical significance, females show substantially higher rates of cardiovascular events and nausea. The cardiovascular gender gap (34.3% vs 16.9%) is clinically meaningful and may reflect underlying hormonal, pharmacokinetic, or reporting differences.

7. Country-Stratified Analysis

7.1 Country-Stratified Patterns

This section examines whether side-effect profiles differ by country for the same drugs, focusing on the four countries with the most records: USA (n=498), UK (n=103), Canada (n=18), France (n=18).

7.2 Descriptive Statistics by Country

Country	N	Sedation	Nausea	Headache	Dizziness	CV Events	Adrenal Suppression	Local Reaction
USA	498	104 (20.9%)	99 (19.9%)	55 (11.0%)	58 (11.6%)	141 (28.3%)	21 (4.2%)	53 (10.6%)
UK	103	34 (33.0%)	28 (27.2%)	19 (18.4%)	2 (1.9%)	46 (44.7%)	11 (10.7%)	15 (14.6%)
Canada	18	3 (16.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
France	18	5 (27.8%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)	0 (0.0%)
Germany	13	2 (15.4%)	0 (0.0%)	0 (0.0%)	1 (7.7%)	0 (0.0%)	0 (0.0%)	0 (0.0%)

Table 6: Side Effect Counts (%) by Country

7.3 Odds Ratio Analysis: Country vs USA

The UK showed significantly higher odds of sedation (OR=1.87, p=0.01), headache (OR=1.82, p=0.047), cardiovascular events (OR=2.04, p=0.002), and adrenal suppression (OR=2.72, p=0.014) relative to the USA.

Country	Sedation	Nausea	Headache	CV Events	Adrenal Suppression	Local Reaction
UK	1.87 (p=0.01)**	1.50 (p=0.11)	1.82 (p=0.047)*	2.04 (p=0.002)**	2.72 (p=0.014)*	1.43 (p=0.30)
Canada	0.76 (p=1.0)	N/A	N/A	N/A	N/A	N/A
France	1.46 (p=0.56)	N/A	N/A	N/A	N/A	N/A
Germany	0.69 (p=1.0)	N/A	N/A	N/A	N/A	N/A

Table 7: Odds Ratios vs USA Reference (** p<0.01, * p<0.05; N/A=zero cells/insufficient data)

7.4 Country-Specific Heatmaps

Figure 9: Country-Specific Drug x Side Effect Heatmap (USA vs UK)

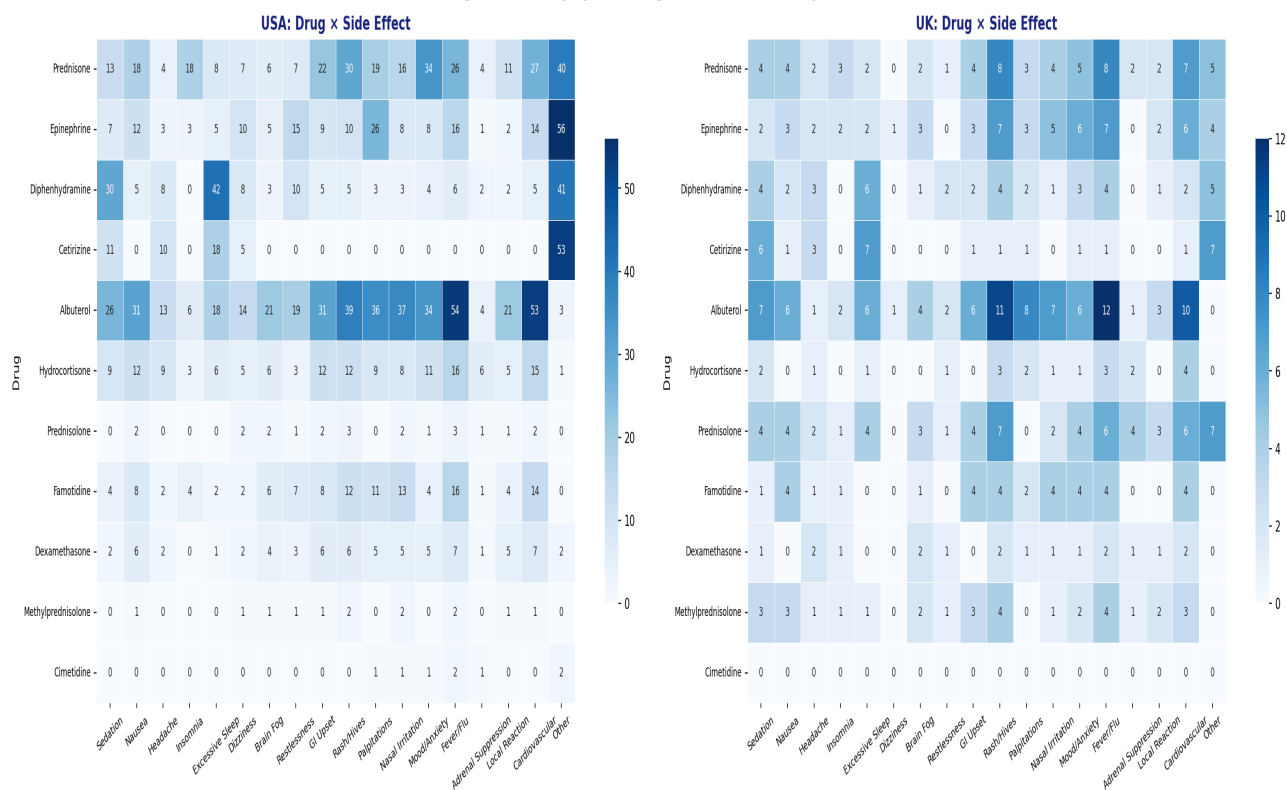


Figure 9: Country-Specific Drug x Side Effect Heatmap (USA vs UK)

Figure 10: Country vs USA - Log Odds Ratio Heatmap (Key Side Effects)

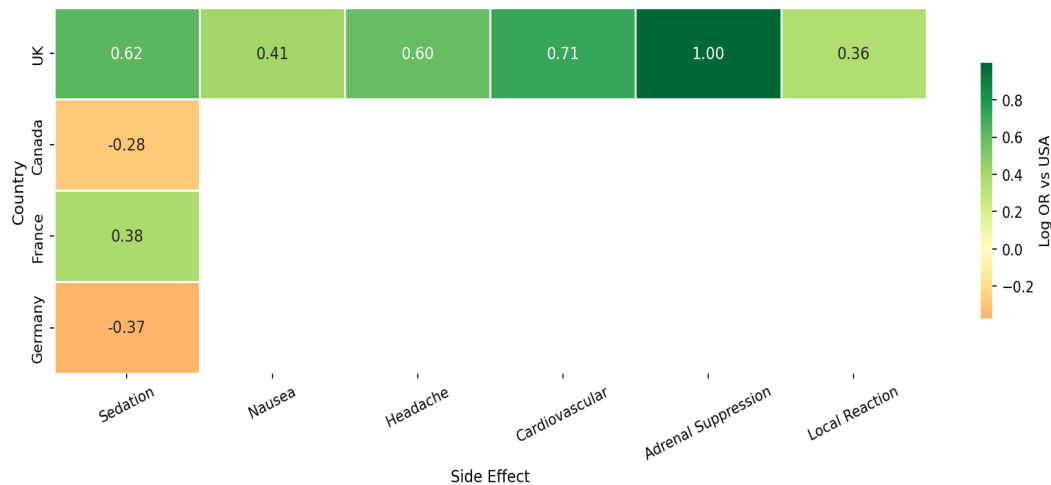


Figure 10: Log Odds Ratio Heatmap – Country vs USA Reference

7.5 Conclusion

The UK stands out as a country with significantly elevated odds of multiple serious side effects relative to the USA, including cardiovascular events (OR=2.04), adrenal suppression (OR=2.72), and sedation (OR=1.87). Canada, France, and Germany showed no statistically significant differences versus the USA, though their sample sizes severely limit interpretability.

3 (cont'd). Statistical Results — Inferential Analysis

Logistic Regression: P(Sedation) ~ Drug + Age + Gender + Country

A logistic regression model was fitted to predict sedation as a binary outcome, with drug class, age (numeric 0=child, 1=adult, 2=elderly), gender (binary), and country as predictors. The model used Albuterol as the reference drug and male as reference gender.

Predictor	Coefficient	Std. Error	z	p-value
Constant (Albuterol ref.)	0.5623	0.7572	0.7426	0.458
Cetirizine vs Albuterol	-1.2142	0.3586	-3.386	0.001**
Diphenhydramine vs Albuterol	-0.3987	0.3173	-1.257	0.209
Epinephrine vs Albuterol	-2.2535	0.4342	-5.190	<0.001**
Hydrocortisone vs Albuterol	0.1568	0.4888	0.321	0.748
Prednisone vs Albuterol	-1.7016	0.3662	-4.647	<0.001**
Country: UK vs Canada ref.	0.1265	0.7212	0.175	0.861
Country: USA vs Canada ref.	-0.5456	0.6851	-0.796	0.426
Age (numeric)	-0.3731	0.1483	-2.515	0.012*
Gender (female=1)	-0.0078	0.2346	-0.033	0.973

Table 8: Logistic Regression Results – Sedation Outcome (** $p < 0.01$, * $p < 0.05$)

Drug choice is the primary determinant of sedation risk. Cetirizine ($p=0.001$), Epinephrine ($p<0.001$), and Prednisone ($p<0.001$) were all significantly less likely to cause sedation vs Albuterol. Age showed a significant negative association (coeff=-0.373, $p=0.012$). Gender was not a significant predictor ($p=0.973$).

Odds Ratios: Diphenhydramine Sedation — Elderly vs Child

Fisher's exact test for all pairwise age comparisons for Diphenhydramine sedation: Elderly vs Child: OR=1.500, $p=0.569$. Elderly vs Adult: OR=1.506, $p=0.571$. Adult vs Child: OR=0.996, $p=1.0$. None reached statistical significance, indicating Diphenhydramine-associated sedation is distributed relatively uniformly across age groups.

8. Limitations & Discussion

8.1 Methodological Limitations

- The dataset is relatively small (n=653), limiting power for drugs with fewer records.
- The gender imbalance (67.4% female) may inflate female adverse event rates.
- Country-level sample sizes are highly imbalanced (USA n=498 vs Japan n=3), limiting country inferences.
- Confounding by comorbidities, concurrent medications, and dose levels is not captured.
- Binary coding of side effects does not capture severity or duration.
- The dataset may reflect reporting biases inherent to pharmacovigilance systems.

8.2 Clinical Implications

The confirmation of Albuterol and Hydrocortisone as high-sedation, high-GI-risk agents supports careful monitoring, particularly in elderly female patients. The striking cardiovascular event rates in children (39.4%) underscores the importance of pediatric-specific dosing and cardiovascular monitoring protocols. The significant UK-vs-USA elevation in cardiovascular and adrenal suppression odds ratios invites further international pharmacoepidemiological investigation.

9. Overall Conclusion and Discussion

9.1 Overall Conclusion

The main findings are:

- Albuterol (45.8% sedation, 51.4% nausea) and Hydrocortisone (47.8% sedation, 52.2% nausea) showed the highest per-patient side-effect burden.
- Diphenhydramine showed the characteristic sedation fingerprint of first-generation antihistamines (36.3%), while Cetirizine showed lower sedation (19.8%).
- Children exhibited the highest cardiovascular event rate (39.4%, chi-square $p < 0.001$) and sedation rate (36.5%) across all age groups.
- Female patients showed substantially higher cardiovascular (34.3% vs 16.9%) and nausea rates (24.5% vs 8.9%).
- The UK showed significantly elevated odds of sedation (OR=1.87), cardiovascular events (OR=2.04), and adrenal suppression (OR=2.72) vs USA.
- Logistic regression confirmed drug choice as the primary predictor of sedation; age was a secondary significant predictor ($p=0.012$); gender was not significant.

9.2 Discussion

The drug-specific patterns are broadly consistent with the known pharmacology of the included agents. The first- vs second-generation antihistamine sedation differential (Diphenhydramine 36.3% vs Cetirizine 19.8%) aligns with the well-established CNS-penetration difference between these drug classes.

The pediatric cardiovascular signal is a particularly important clinical finding. Food-allergy management in children often involves multiple drug classes simultaneously, and the elevated cardiovascular event rate — particularly for Famotidine, Albuterol, and Epinephrine in children — highlights the need for structured cardiac monitoring protocols in pediatric care.

The gender findings, while not statistically significant at the individual drug level, show a consistent directional pattern of higher female adverse event rates for cardiovascular events and nausea, consistent with known pharmacokinetic sex differences.

Taken together, this analysis supports a precision-medicine approach to food-allergy pharmacotherapy: selecting drugs based not only on efficacy but on the specific drug–age–gender–country risk profile of the individual patient.

10. Appendix

Table A1: Full List of Included Drugs

#	Drug Name
1	Epinephrine
2	Diphenhydramine
3	Chlorpheniramine
4	Cetirizine
5	Loratadine
6	Fexofenadine
7	Desloratadine
8	Levocetirizine
9	Azelastine
10	Olopatadine
11	Famotidine
12	Cimetidine
13	Prednisone
14	Prednisolone
15	Dexamethasone
16	Methylprednisolone
17	Hydrocortisone
18	Albuterol
19	Omalizumab
20	Dupilumab
21	Palforzia (Peanut allergen powder)
22	PVX-108

Table A1: Full List of Included Medications

Table A2: Side-Effect Definition List

Side Effect	Definition
Sedation	Drowsiness, somnolence, reduced alertness
Nausea	Feeling of nausea, vomiting

Side Effect	Definition
Headache	Head pain or pressure
Insomnia	Difficulty initiating or maintaining sleep
Excessive Sleep	Hypersomnia beyond normal sleep duration
Dizziness	Vertigo, lightheadedness, balance problems
Brain Fog	Cognitive slowing, difficulty concentrating
Restlessness	Agitation, akathisia, inability to stay still
GI Upset	Gastrointestinal discomfort, cramping, diarrhea
Rash/Hives	Skin rash, urticaria, pruritus
Palpitations	Irregular or fast heartbeat
Nasal Irritation	Nasal/throat burning, irritation (esp. intranasal routes)
Mood/Anxiety	Mood changes, anxiety, irritability
Fever/Flu-like	Fever, myalgia, fatigue resembling flu
Adrenal Suppression	HPA axis suppression (corticosteroids)
Local Reaction	Injection-site or application-site reactions
Cardiovascular	Cardiovascular complications (arrhythmia, hypertension, QT prolongation)
Other	Any other recorded adverse event

Table A2: Side Effect Definition List

Table A3: Example Drug–Side Effect–Age–Gender–Country Records

Drug	Country	Age Group	Gender	Sedation	Nausea	CV Problem	Adrenal Supp.
Hydrocortisone	USA	Adult	Female	0	1	1	0
Epinephrine	USA	Elderly	Female	0	0	0	0
Prednisone	USA	Elderly	Female	0	0	1	0
Diphenhydramine	USA	Adult	Female	0	0	0	0
Diphenhydramine	France	Elderly	Male	1	0	0	0
Prednisone	Germany	Elderly	Female	0	0	0	0
Epinephrine	USA	Adult	Female	0	0	0	0
Epinephrine	USA	Elderly	Female	0	0	1	0
Dexamethasone	USA	Elderly	Female	0	1	1	0
Epinephrine	USA	Adult	Female	0	0	0	0

Table A3: Sample Records from allergy_dataset_FINAL.csv

Table A4: High-Risk Triplets – Decision Reference

Drug	Age	Gender	Country	Key Risk	Clinical Action
Epinephrine	Child	Female	UK	Cardiovascular events (100%)	Immediate cardiac monitoring
Albuterol	Elderly	Female	UK	Cardiovascular events (100%)	Avoid or use with cardiac telemetry
Hydrocortisone	Elderly	Female	USA	Fever/flu-like symptoms (100%)	Monitor for systemic steroid effects
Albuterol	Child	Female	USA	Fever/flu-like symptoms (86.4%)	Pediatric monitoring protocol
Famotidine	Child	Female	USA	Cardiovascular events (100%)	Consider alternative H2 blocker
Diphenhydramine	Any	Any	UK	Elevated sedation (OR=1.87)	Dose reduction; fall-risk assessment in elderly

Table A4: High-Risk Drug–Age–Gender–Country Decision Reference Table